Oliveres *et al. Fire Ecology* (2025) 21:9
https://doi.org/10.1186/s42408-024-00348-x

Fire Ecology

ORIGINAL RESEARCH

Open Access

The path to strategic fire management planning in the Aran, Pyrenees

Jordi Oliveres[1,2*], Marc Castellnou[2,3], Xavi Castellarnau[2], Martí Rosas-Casals[1], Lluís Brotons[4,5,6] and Andrea Duane[3,4,7]

Abstract

Background This paper aims to present insights about the beneficial use of fire within a strategic and operational framework, providing actions for better adaptation to fire regime shifts in the face of climate change and land use changes. Supported by transdisciplinary analysis and aimed at supporting an actual regulation in the Aran (~ 600 km^2 in the Spanish Pyrenees), our case study focuses on creating a management path toward landscape resilience at the social and ecological level.

Results We present a novel approach for fire management in Europe, where fire is no longer treated as an isolated, unpredicted, and unwanted element but where the fire regime (considering both wildfires and prescribed burns) is managed as a whole and under changing conditions. Under this approach, some unplanned ignitions represent a window of opportunity where emergency responders can turn wildfire incidents into land management goals. This way, the present framework helps to reduce uncertainty by proactively anticipating plausible future scenarios. In addition, the subjacent strategic innovation dissolves the trap linked to fully aggressive fire suppression strategies, proposing a transformative role for emergency management organizations to overcome the burden of extinction everywhere, every time.

Conclusions This work highlights the importance of understanding fire complexity through transdisciplinary knowledge, acknowledging its historical significance for rural populations, and recognizing fire as a cultural heritage and essential ecological process that shapes the landscape. The study expects to serve as a catalyst for fire resilience in the region and inspire other mountainous areas to address similar challenges posed by global change.

Keywords Environmental flow of fire, Fire management, Fire polygons, Fire resilience, Prescribed fire

Resumen

Antecedentes Este trabajo tiene como objetivo presentar distintas perspectivas sobre el uso beneficioso del fuego dentro de un marco estratégico y operacional, proveyendo de acciones para una mayor adaptación a los cambios en los regímenes de incendios en el contexto del cambio climático y de cambios en el uso de la tierra. Basado en un análisis transdisciplinar y con el objetivo de apoyar una regulación real en el Arán (~ 600 km^2 en los Pirineos de España), nuestro estudio de caso se enfoca en crear un marco de manejo orientado a lograr la resiliencia del paisaje tanto al nivel ecológico como social.

Resultados Presentamos una aproximación novedosa para el manejo del fuego en Europa, donde el fuego no se trata como un elemento aislado, impredecible y no deseado sino en el cual, el régimen de fuego (considerando tanto

*Correspondence:
Jordi Oliveres
itbop317@gencat.cat
Full list of author information is available at the end of the article

© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

incendios como quemas prescritas) se maneja como un todo y bajo condiciones cambiantes. Bajo este enfoque, algunas igniciones no planificadas se presentan como una ventana de oportunidad cuando quienes responden a las emergencias pueden transformar incidentes de incendios en objetivos de gestión del territorio. De esta manera, el presente marco de trabajo ayuda a reducir las incertidumbres de forma proactiva anticipando escenarios plausibles a futuro. Adicionalmente, la subyacente innovación estratégica disuelve la trampa ligada a las estrategias de supresión totalmente agresivas, proponiendo un rol transformador para las organizaciones de gestión de emergencias superando la insostenible carga de la extinción en todo lugar y en todo momento.

Conclusiones Este trabajo subraya la importancia de entender la complejidad de los incendios a través del conocimiento transdisciplinario, aceptando su significado histórico para las poblaciones rurales, y reconociendo al fuego como una herencia cultural y un proceso ecológico esencial que modela el paisaje. Este estudio espera que sirva de catalizador para lograr la resiliencia del fuego en la región e inspire a otras áreas montañosas para enfrentar desafíos similares que impone el cambio global.

Introduction

Changes in wildfire regimes are a global concern, and Europe is no exception. Landscape dynamics are shaped by the accumulation of fuels—primarily due to rural abandonment—an increase in wildland-urban interfaces and climate conditions that dry out fuels and extend burning seasons. These factors contribute to a latitudinal and altitudinal variation in wildfire occurrences (de Rigo et al., 2017). The Mediterranean basin is particularly vulnerable to these trends, where the effects of wildfires are worsening (Alvarez et al. 2024), and fire represents a threat to human populations, especially in its mountainous areas (Cramer et al. 2020).

Increasing the number and capabilities of response units and a strong emphasis on fire suppression is a common policy adopted by public administrations to address the growing threat of wildfires. While upgrading these capacities may be adequate in many instances, the trend toward fully aggressive suppression strategies in Europe is an unsustainable management practice since there is a physical limit to the extinction capacity, particularly during extreme wildfire events (Tedim and Amraoui 2018; Castellnou et al. 2019; Duane et al. 2021). The effectiveness of fire suppression in the short- and medium-term might favor large wildfires through a widespread increase in fuel amount and continuity (Minnich 1983; (Fry et al. 2014; Schoennagel et al. 2017). Moreover, if larger fires become more frequent due to suppression policies and climate change, this can lead to landscape homogenization, creating a positive feedback loop that exacerbates the problem (Loepfe et al. 2010). Overall, a total fire exclusion policy is far from being a neutral process at the socio-ecological level, potentially contributing to, expanding, and intensifying the problem of wildfires beyond forest systems (Kreider et al. 2024; Xanthopoulos et al. 2020).

Given this situation, a paradigm shift toward fire resilience is necessary (Newman Thacker et al. 2023), along with leveraging solutions for socio-ecological transformations (Moreira et al. 2020; Tölgyesi et al. 2022; Copes-Gerbitz et al. 2022; Buisson et al. 1979). This shift goes beyond analytical methods and technology-based solutions; it advocates for a policy that (a) recognizes the ecological role of fire as a natural process and cultural heritage and manages the fire regime as a whole (Duane et al. 2021) and (b) strengthens interconnections between landscape management policies and scientific knowledge (Fernandes 2013).

In Europe, especially in the Mediterranean, fire has been often advocated and has also been used traditionally in land management for millennia (Bal et al. 2011). However, at present, its use is scarce and responds to (1) cultural burning (known as traditional fire use in the European context) (2) prescribed burning, and (3) fire operations to fight wildfires (Pastor et al. 2009; Montiel 2010; Fernandes et al. 2013). A majority of expert voices advocate for an increase in the use of prescribed fire to reduce fuel accumulation and decrease wildfire risk at the landscape level (Montiel 2010; Rego et al. 2010; Duane et al. 2019; Davim et al. 2022), as well as ecosystem management and biodiversity conservation (Sande 2010; Fernandes 2022). Nonetheless, prescribed burning is still far from representative of business as usual in Europe (Fernandes 2022).

In the Pyrenees, the mountain range straddling the border of France and Spain, fire activity is not a new or anecdotal phenomenon but a historical one (Bal et al. 2011; Fernández-Giménez 2015; Rius et al. 2012). Fire use was a regular management tool for shepherds in this region when the Pyrenees fully established transhumance (Messana et al. 2023; Coughlan 2014). Since the beginning of suppression policies in Catalonia (NE Iberian Peninsula) in the 1980s, the traditional role of

fire in managing landscapes has been controversial (Otero and Nielsen 2017). Traditional fires in the Pyrenees mainly respond to extensive regeneration of pastures and control of pastures encroaching into shrubs. With a few exceptions, these pastoral burns were ruled out in the 1980s when policymakers developed highly restrictive environmental laws in response to large wildfires occurring in other parts of the country (Llei 6/1988, de 30 de març, forestal de Catalunya). The prohibition of burning imposed by regulation enforcement has generated disaffection among land managers of the Pyrenees, and occasionally, it has led to uncontrolled ignitions that ultimately have required the deployment of fire responders. Additionally, many requests for controlled burns are not approved due to logistical or bureaucratic reasons. This results in contradictory situations where wildfires occur in areas where fire historically had socioecological benefits—and may even have a prescribed burning plan—yet are still subjected to direct suppression. This situation increases the vulnerability of emergency management organizations (EMO), heightens frustration among managers, and undermines the socioecological benefits associated with the traditional use of fire. In addition, the Pyrenees mountain range is expected to increase fire behavior in the following decades due to a decrease in fuel moisture linked to climate change (Resco de Dios et al. 2021). In this context, the business as usual of fully aggressive suppression in the Pyrenees needs a political change toward landscape resilience (Newman Thacker et al. 2023) that includes democratization of risk consensus (González-Hidalgo et al. 2014; Otero and Nielsen 2017), the integration of the current knowledge available from local land managers, fire experts, and scientific communities, and the coexistence of interests from stakeholders of the socioecological system.

Arising from the need to bridge the gap between the constraints and challenges above, we present here a new strategic framework to minimize wildfire-induced damage and scale the positive impacts of fire use in the Aran Valley, Pyrenees (NE Iberian Peninsula). We also present the first case in which this framework was fully implemented. Using a transdisciplinary approach, our proposal creates new fire response scenarios by prioritizing the common goals shared between the local actors and stakeholders. The main objectives related to co-creating this strategic framework for fire management were (1) to decrease the uncertainty derived from uncontrolled ignitions, (2) to improve emergency organizations' capacity to deal with simultaneity, and (3) to enhance the positive effects of fire use at a landscape scale.

Region description

The Aran Valley is a Pyrenean valley of 633.6 km^2 with around 10,268 inhabitants (2022) located in Spain and bordering France (Fig. 1). Administratively, the Aran is a "unique territorial entity" with additional administrative powers compared to the rest of the region (Catalonia, Spain). This status was most recently formalized in February 2015 (Llei 1/2015 del 5 de febrer, del règim especial d'Aran). However, the Aran has had an autonomous government, with specific responsibilities in environment

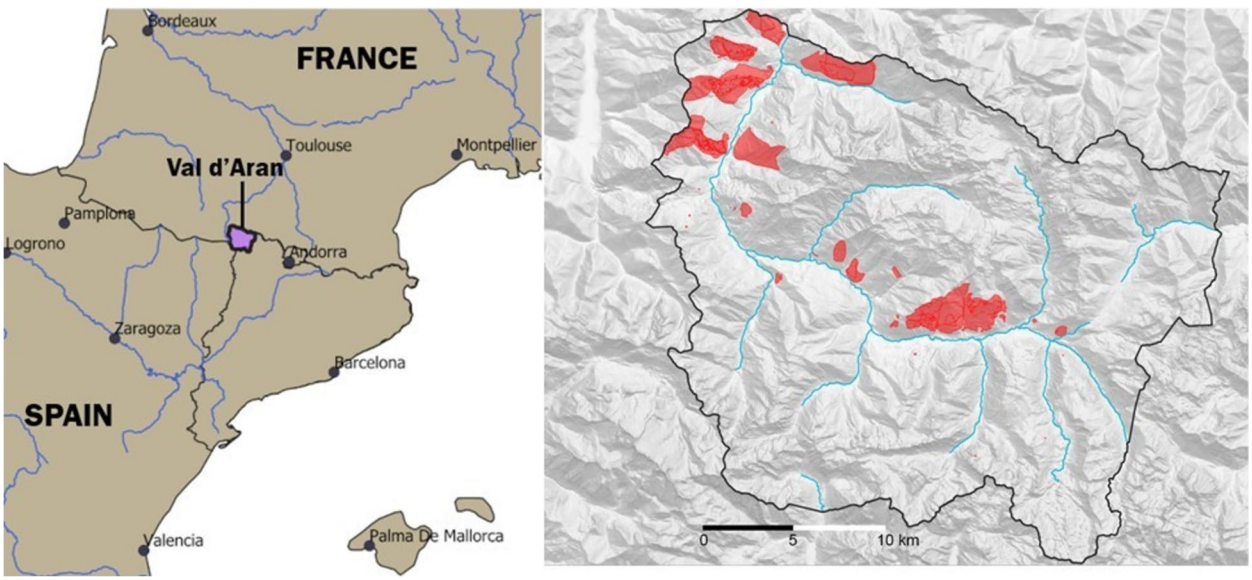


Fig. 1 The left panel shows the Aran (purple) between the NE Iberian Peninsula and France. The right panel shows the relief (gray), rivers (blue), and occurrence of fire since 1990 (red) in the Aran (black border)

and emergency policies, since 1991 called the General Council of the Aran. Additionally, land ownership in the Aran is mainly composed of communal lands, which tend to occur at higher elevations and reduce the complexity of any attempt to implement formal planning at a landscape scale.

The Aran is part of the Atlantic River basin (i.e., rivers flow north, towards France) and is in the central Pyrenees, where the mountain range's significant altitudes are found. This mountainous area has a mean altitude of 1840 m (from 578 to 2418 m, with lower altitudes in the region's northern part) and a mean slope of 27°.

The Aran climate is mostly Atlantic, which makes it unique in the surrounding Catalan context. Annual precipitation and average temperature range from 900 to 1200 mm (with maximum precipitation peaks in summer, usually due to convective storm activity) and from 6 to 10 °C, respectively. Wind is highly variable (with predominant wind directions being south, northwest, and northeast), and it fluctuates spatially mainly because of relief and altitudinal changes. Evidence from the last decades shows that temperatures are rising at a 0.2 °C per decade rate (1959–2010), especially in summer months, and precipitation is decreasing at a 1.5% per decade rate (SMC, 2023). Climatic projections agree on a warming intensification in the area (Amblar-Francés 2020). Approximately 35% of the region is covered by grasslands and mountain pastures (in low, medium, and high elevations), 17% by shrublands, 32% by forests (in lower and mid elevations), and the remaining 16% by (in decreasing order) rocky areas, agricultural fields, urban areas, and water bodies (González-Guerrero and Pons 2020).

The fire rotation period (Arno 1994) for the entire region is about 350 years. This value must be interpreted cautiously because it accounts for wildfires registered in the last decades but ignores historical fire use. Pastoral burns were common in the area in the recent past (last century) and used extensively to manage the Aran's open landscapes, as local rangers and farmers report. According to the official fire perimeter data sourced from the Catalan Government agency responsible for wildfire prevention (Servei de Prevenció d'Incendis de la Generalitat de Catalunya) together with the CFRS data source, 66 fires have been reported from 1990 to 2017, affecting 5062 ha with an average of 180 ha per year. In this period, the 20 largest fires (30% of the total number of fires) burned up to 86% of the total burnt area, with the number of fires and burnt areas peaking in the winter season (January, February, and March). Reports from the Pyrenees Observatory (OPCC) support that the landscapes in this part of the Pyrenees are green and humid in a typical summer. In contrast, in winter, low temperatures typically inhibit vegetative activity, and fires propagate below the snow altitude level when vegetation is drought-stressed and environmental moisture is low. Organic soil depth also plays a relevant role in this fire regime, like in Boreal or Euro-Siberian regions: suppression efforts become harsher when the organic soil layer (usually quite deep) is dry enough and becomes available to burn. This poses more difficulties in suppressing fires, which can burn for more days if soil fire has not been mopped up properly. When the peat fire is still smoldering, fire can spread again under changing weather conditions, affecting new areas.

Fires in the Aran mainly propagate according to two major spread patterns (Duane et al. 2016; Costa 2011): topographic fires that follow southern slopes during the day and usually stop when slope and aspect change, and wind-driven fires that interact with the complex topography. The latter usually have a higher rate of spread when converging with slope or wind, generate medium–high intensity with several meters of flame lengths, and, depending on dry conditions, may have a null, partial, or total capacity to penetrate and consume the organic soil. The intensity and severity are considered high when they involve and consume the organic soil layer (ground fires) or when tree canopies are affected (crown fires). In contrast to southern aspects, north-facing slopes have lower fire activity. They are usually dominated by deciduous and coniferous forests, a mosaic of southern and northern environments providing a rich heterogeneity at the landscape scale (González-Guerrero and Pons 2020).

An active fire suppression-based strategy has been applied in the region since the 1980s. The administration occasionally implements prescribed burns to inhibit uncontrolled ignitions. However, this objective is only partially with some contradictions such as suppressing a fire in a place where the administrative level has been committed to implementing prescribed burns shortly (Fig. 2). Moreover, considering that teams involved in suppression and prescribed burning are the same, it implies that fire responders assume extra risks and double responsibility.

Strategic framework

This section details the process and result of the strategic fire use planning process and results in the Aran. First, we introduce the conceptual framework and process design involved in strategic planning (the "Conceptual framework and process design" section). Then, we present the new fire management plan tailored to the Aran and the first case in which this strategic loop was fully



Fig. 2 Fire suppression operations (**A**, **B**, **C**) and prescribed burn operations (**D**, **E**) just a few weeks after the fire in the same specific area (Garòs wildfire, 2017)

Source: Catalan Fire and Rescue Service

implemented (the "The fire management plan for the Aran" section).

Conceptual framework and process design

The conceptual framework presented here builds on the view that, in many cases, some wildfires can be seen as suppliers of ecosystem services (Pausas and Keeley 2019), and they can play a role in management strategies aimed at fostering landscape resilience at social and ecological levels (Brotons 2013; Otero and Nielsen 2017; Duane et al. 2019; Pausas 2019; Moreira et al. 2020; Newman Thacker et al. 2023). From this mindset, we propose the fire resilience paradigm (Fig.3). In this mindset, climate change and land use change act as drivers that force changes in pyro-climatic niches. The intensity and severity of wildfires are correlated with the spatial and temporal scale of these changes, which, at the same time, forces an adaptation process for EMO. Here, an important balancing feedback loop exists between EMO's adaptability and wildfire intensity and severity, which passes through the definition and creation of resilient landscapes. Conceptual and strategic frameworks like the one detailed here exploit the intrinsic complexity of fire and landscape dynamics in favor of fire resilience at both ecological and social levels.

Table 1gathers fundamental fire-related concepts to understand the roots of the process design. These concepts embrace transdisciplinary knowledge from scientific research and fire experts (Beans & Evans 2023; Arilla 2023; Castellnou et al. 2019; FMB 2019). In addition to Table 1, we propose the concept of the environmental flow of fire.

Environmental flow of fire (EFF)

This concept is an outcome of this research that interconnects fire-theoretical related concepts with the operational mindset of fire practitioners and fire responders. As a metric for supporting decision-making, it has been inspired by the field of water management, more specifically by the concept of environmental flow (Tickner et al. 2017; Scanlon 2003). *EFF is the ideal amount of fire in a landscape required to preserve the common goals formally stated for a given socio-ecological system.* We must highlight that for the environmental flow of fire, we include fire implemented as prescribed fire but also fire managed during

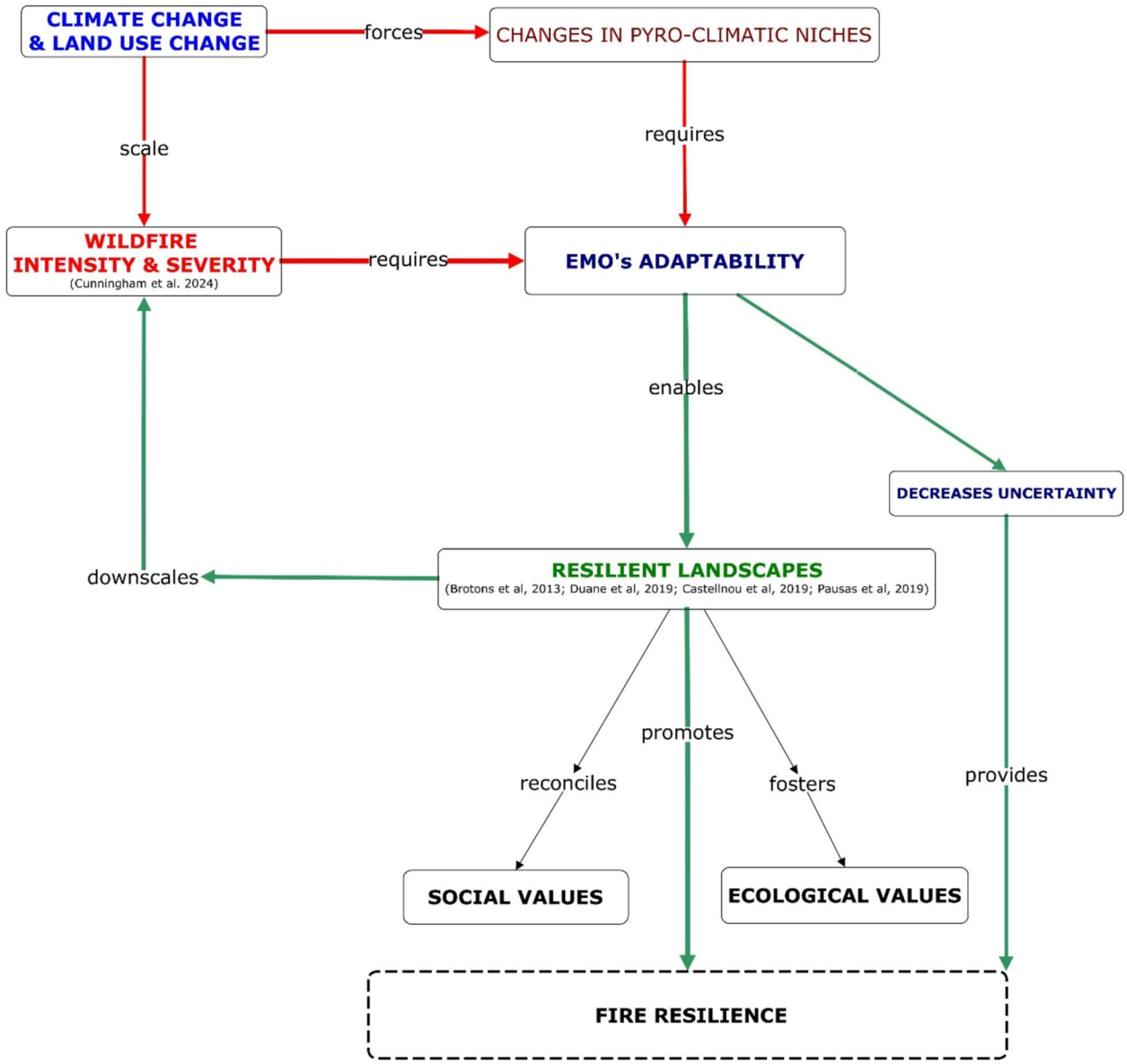


Fig. 3 Concept map linked to a fire resilience paradigm. EMO stands for emergency management organizations

wildfire incidents when outlined in land management plans.[1]EFF has connotations of public benefit since common goals, particular to each socio-ecological system, become shared priorities that must be recognized in formal planning and considered in emergency management decision-making. All common goals must align with fire responders' safety, civil protection improvements, and ecological processes supporting healthy forest ecosystems and multifunctional management. EFF responds to a set of components that are similar to fire regime characterization. These components are (fire use) capacity (CA), (fire use) power (P), fire behavior (B), fire seasonality (S), and fire frequency (R). These components should be easy to incorporate for managers and decision-makers, except for capacity and power, which need to be estimated. We omit the definitions of fire behavior, seasonality, and frequency in this manuscript because all are widely described in

[1] This strategy of fire management during wildfire incidents outlined in land management plans has been nonexistent in Europe due to the idiosyncrasy of European landscapes and societies, with densely populated areas, few remote natural sectors, a prevalent private landownership regime, and a historical policy of total fire exclusion. Prescribed natural fire, wildland fire use, and managed fire are archaic concepts related to the use of wildland fire. Those names respond to similar concepts that began to evolve formally in the United States in 1968 and developed primarily in national parks with non-human exposition and naturally ignited fires (Wagtendonk, 2007).

Table 1 Conceptual definitions

Emergency pre-planning (EPP): A strategic agreement pre-approved by the community and designed to anticipate resolution scenarios before emergencies. This strategy enables optimal triage to prevent collapse and preserve the common goals at a socio-ecological scale (Castellnou et al. 2019)
Fire management unit (FMU): A defined land management area characterized by specific objectives, topographic features, access, values to be protected, political boundaries, fuel types, and major fire regimes that differentiate it from adjacent units. FMUs may have dominant management objectives along with preselected strategies to achieve these goals
Fire management area (FMA): A sub-geographic area within a fire management unit (FMU) designated as a predefined acceptable management zone for a wildland fire managed for resource benefits. Predefinition of these areas facilitates planning for fire management, identifying threats to life, property, resources, and boundaries, and identifying initial actions
Fire management plan (FMP): A strategic plan that defines a program to manage wildland and prescribed fires and documents the program in the approved land use plan. Fire management is the process of planning, preventing, and fighting fires to protect people, property, and forest resources. It also involves using fire to attain forestry, wildlife, and land-use objectives
Fire polygons: Although different products refer to fire polygons (Arilla et al. 2023), in this work they are defined as polygons of fire potential, which consider two simultaneous elements for their dimensioning: fire behavior and the work opportunities for firefighters. Thus, the contour used to delimit a polygon of fire potential represents a location of the territory where changes in the fire behavior may occur, which at the same time represents work opportunities for fire suppression
Prescribed fire (Rx): Any fire ignited to meet specific management objectives. Prior to ignition, a written, approved, prescribed fire plan must exist, and legal requirements must be met in coherence with the existing law

the bibliography (Agee 1998; Arno 1994) and are well-known by fire experts. However, we propose to categorize fire behavior in ground, surface, torching, crown fires categories, and seasonality variables into spring, summer, autumn, and winter seasons to simplify decision-making during emergency management. Power (P), measured in hectares/day or hectares/episode, is the maximum manageable area within an FMA in a single incident or event. Capacity (CA) responds to the polygon representing the maximum manageable area. CA is measured in hectares. It equals the size of an FMA; therefore, the value is null for any ignition outside them. Power and capacity do not respond to specific formulas, but both components are the result of integrating environmental management conditionings and applying tactical analysis, according to public land managers and local fire responders.

The process design for the strategic framework results from combining two methods: (1) an approach based on fire expertise and social participation and (2) the results of a landscape dynamic modeling exercise built for the Aran Valley, including different fire management scenarios. Since this paper focuses on the conceptual framework, the model description has been omitted but can be accessed in (Duane 2020).

First, the General Council of the Aran Valley developed a participation process at the municipality level, engaging their representatives in the decision-making process linked to the strategic framework. It pursued four main objectives: (1) to share the goals aimed at strategic innovation in fire management, (2) to gather the specific worries concerning wildfires, (3) to gather the specific demands for pastoral burns, and (4) to identify the common goals shared by the leading actors officially represented in the Aran, including public land managers, non-lucrative associations, among others. As a result, the common goals and shared priorities for the Aran's socioecological system were stated according to the participative process:

1. Prevent high-severity impacts at ecological and social levels derived from large, high-intensity fires.
2. Enhance emergency organizations' resilience (to avoid collapse in the face of simultaneity, thanks to the possibility of strategic triage).
3. Promote extensive livestock farming by combining grazing animals and fire use.
4. Enhance biodiversity conservation at the landscape scale (patches of extensive open habitats within a dominant tree forest matrix). The primary objective of fire deals with pasture regeneration and extensive scrub population control within the open habitats.
5. Foster biodiversity conservation at the habitat scale (different ages of regeneration thanks to different fire disturbances implemented in a prescribed manner).
6. Prevent geological risks, such as large avalanches and landslides, from the impacts of high-intensity fires in protective forests.
7. Enhance open habitats for hunting resource management.
8. Preserve the heritage inherent to the Aran's landscapes as a sustainable economic-touristic resource.

These common goals and shared priorities provided consensus and strategic direction, including identifying current stakeholders' demands. Additionally, this step was intended to facilitate socialization in emergency

management decision-making (Otero et al. 2018; (Otero and Nielsen 2017) and more intelligent resource allocation in fire response.

In parallel, the Catalan Fire and Rescue Service (CFRS) developed the forest fire potential polygons (Arilla 2023). This zoning procedure is coherent with observed fire behavior in the Aran (fires since 1990) and considers suppression opportunities according to the CFRS's capabilities. The analysis identifies critical points for windy and topographic fire spread patterns, providing the vision to assess the effectiveness of different potential management areas. Those critical points were identified for the entire Aran region. Afterward, we overlaid critical points with historical wildfires to assess current local demands of fire use gathered with the social participation process. This transdisciplinary process ended up in the definition of strategic management areas where fire use is desirable (implementing Rx) or acceptable under specific conditions in case of wildfires (unplanned ignitions). In addition, the CFRS ensured that the tactical and technical details of the plan could be available through a GIS platform for fire managers.

The fire management plan for the Aran

The outcome of the process explained previously is the Strategic Fire Management Plan of the Aran (2022). Original documentation is accessible from Aran's official register (SFMP). The scope of action of the SFMP includes the whole of the Aran, around 63,000 hectares, where we defined areas compatible with fire use (FMAs) and other places where fully aggressive suppression will be systematically applied. With a 10-year horizon since official approval, the SFMP was conceived between 2020 and 2021 and formally approved unanimously (nine municipalities) in May 2022. The impacts of different management alternatives, including suppression at all costs, are gathered in the environmental impact assessment inherent to the legalization process. This information can be consulted in Aran's official register as an appendix of the SFMP.

In terms of results, the core of the SFMP is the set of fire management areas (FMAs). The sizes of the FMAs are notably larger than those typically adopted for the CFRS when implementing prescribed burns in mountain areas. This is because, in the SFMP, we prioritized natural limits in the design of the confinement axes. Robust confinement axes serve as maximum, emphasizing that these management areas serve as a limit rather than a target. On the one hand, the sizes of FMAs help to establish desirable and acceptable strategic scenarios in emergency management decision-making. In addition, the limit of FMA identifies the scenario to avoid (Arilla 2023). Each FMA has a technical sheet on which the EFF (see Table 1) is specified, together with other relevant information such as anchor points (advantageous location, usually a barrier to fire spread, used to reduce the chance of firefighters being flanked by fire). This type of information is critical at a tactical level to contain the fire within the pre-established limits of confinement (Fig. 4). Recall that EFF includes management of wildfire and prescribed fire, which, under this framework, become shared priorities that must be (a) recognized in formal planning since they are particular to each socio-ecological system and (b) aligned with fire responders' safety, civil protection improvements, and ecological processes supporting healthy forest ecosystems and multifunctional management.

The supplementary information offers an example of Canejan's FMA. Such information is also a geodatabase and can be visualized in a GIS platform. Hence, we ensure that geodata is manageable for end-users.

Based on a landscape fire succession model tailored to the Aran Valley under various management scenarios, it was established that 0.66% of the natural land covers should be managed with fire to increase resilience and minimize efforts (Duane 2020). Considering a 10-year horizon, this value implies approximately 7% of the Aran. In practice and depending on different environmental conditions such as snowline levels or fuel availability, the EFF executed within each FMA can oscillate significantly below the maximum manageable scenario. The size of each FMA mainly relates to the physical features of a landscape where fire responders have favorable conditions to contain the spread of a wildfire. That is, the limits and the size of FMAs respond to robust opportunity buffers for fire responders. Regarding the EFF, their values provide the how and when, while the polygon of the FMA determines where and the maximum size. Behavior, seasonality, and frequency variables share the same criteria for all FMAs defined: surface behavior (avoiding ground fires to mitigate adverse impacts concerning soil interaction), autumn and winter seasons (to minimize negative impacts on plants' phenology), and a 7-year frequency (time-effective result concerning critical scrub encroachment processes (see Duane 2020).

Fire use operates at two levels (Fig. 5): use of unplanned ignitions for multifunctional management goals at a landscape scale (hundreds of hectares) and, at a finer scale, planned ignitions in agreement with prescribed burns (tens of hectares). When an unplanned ignition starts to spread within an FMA, it can evolve to fire use only if it is coherent with the emergency preplanning; that is when conditions are compatible with tactical capacity (resources needed are available and positioning is favorable) and coherent with

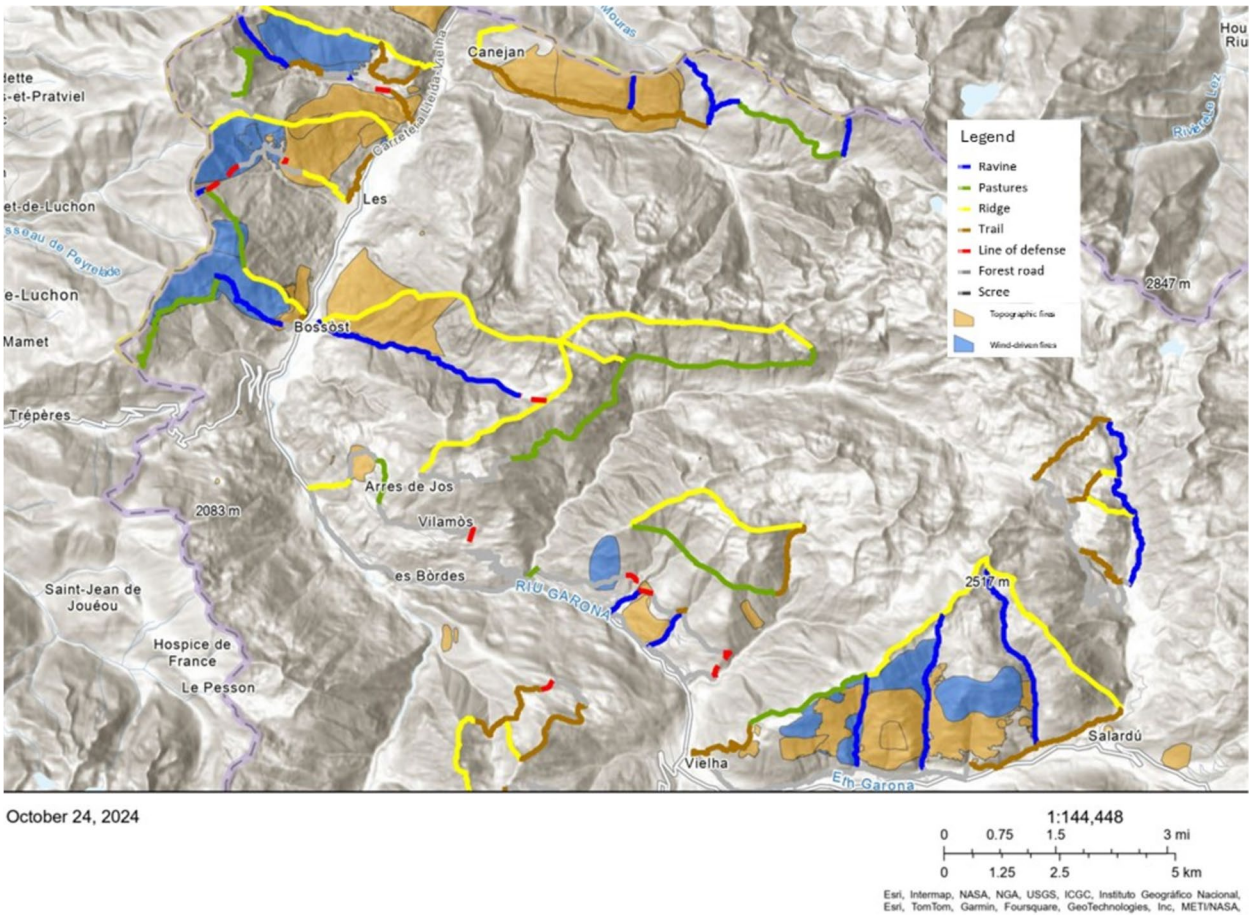


Fig. 4 Types of anchor points as limits of the FMAs. Wind-driven (blue) and topographic fires (orange) since 1990 are also shown. (Credits: authors)

the environmental criteria set out in the plan. This happens regardless of the ignition's cause: the management opportunities are not constrained to "natural" ignition sources but embrace the use of unplanned ignitions caused by sources such as power lines or similar accidental ignitions. With this plan, there is the opportunity to synchronize prescribed burning targets with wildfire operations. Consequently, emergency management organizations can allocate resources to attain pre-approved strategic goals for multifunctional forest management. A fire use strategy different than prescribed fire can only be implemented using procedures linked to an incident command system (ICS). Thus, management of wildfires requires systematically considering operational procedures under an incident action plan (IAP) led by emergency management organizations. The incident commander, assessed by a team of local public managers, integrates management goals stated in the SFMP into decision-making. Consequently, the whole or part of the management goals may be incorporated as tactical objectives within the IAP.

Example of the implementation plan

The SFMP kicked off in 2023 with the CFRS's team of wildland fire specialists, also known as GRAF, leadership. Prescribed burns started in February 2023 (Fig. 6), and the first wildfire incident managed under the SFMP was the Canejan fire in March 2023 (Oliveres 2023). The management of this incident materialized in (1) a direct attack in the left flank to protect the village of Canejan, (2) monitoring the spread of the head at its upper limit (ridge), and (3) accompanying the fire on the tail-right flank, anchoring it at its preplanned lower boundary (Fig.7). We want to highlight these operational details because it is crucial to understand that, from an incident commander's mindset, managing a wildfire is far from a let-it-burn policy. This incident occurred within Hont Hereda (unit A) and Era Tueta (unit B), always remaining within the limits of the FMA polygon. Implementing this strategy required fourteen workdays until final extinction, and it managed a total area of 327 hectares, which falls within the technically/pre-planned manageable area in one event (346 ha). Positive feedback from the Canejan

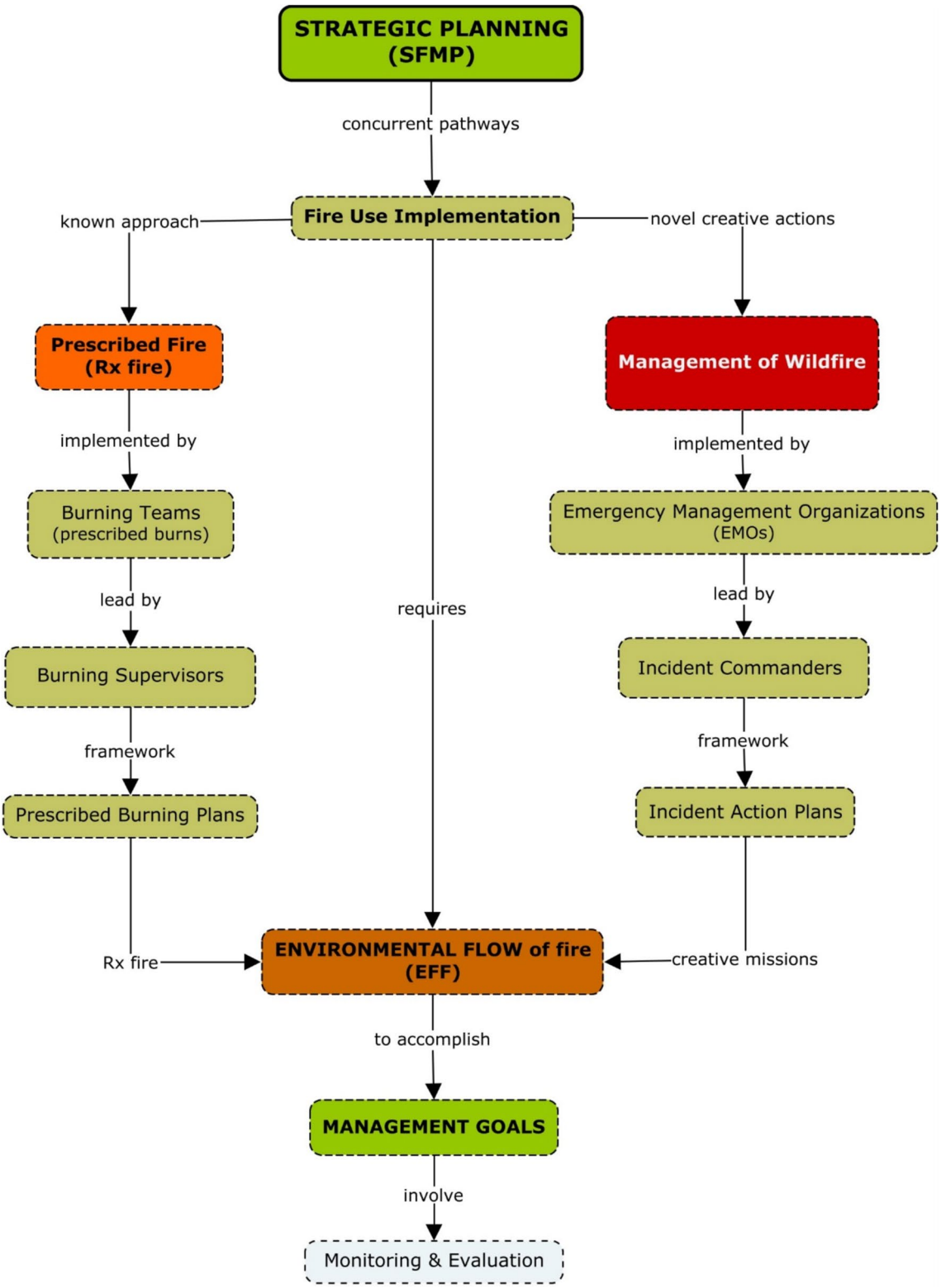


Fig. 5 Illustration of the two concurrent pathways for scaling fire use in the Aran, according to strategic wildland fire management planning



Fig. 6 Prescribed burn in Bossost, Aran (February 2023). Captures **a**, **b**, and **c** illustrate different perspectives of the burn, and capture **d** shows a general view of the unit just after the burn. (Credits: GRAF team, CFRS)

experience is that the capacity of the fire management area provides a tactically credible perspective because the confinement axes, in general, are robust enough to minimize the risk of fire spreading out of the FMA. In addition, the Canejan fire exemplifies the positive non-linear relation between the resources involved and managed goals achieved. The size and qualification of the team that implemented the prescribed burn in Bossost (February 2023) was practically the same as that operated in Canejan a few weeks later (March 2023). Managing planned ignitions in agreement with prescribed burns (tens of hectares) requires the same resources as managing unplanned ignitions for multifunctional management goals at a landscape scale (hundreds of hectares). This organizational performance represents a radical innovation since it produces a 10× improvement when comparing the experience of Canejan (around 300 hectares) with the one of Bossost (around 30 hectares). Also, the possibility of allocating resources on the field varies significantly depending on the situation; dispatching a team to wildfire incidents is less complicated than dispatching the same crew for prescribed burning. Additionally, regarding fire behavior, no significant difference was observed between both experiences; high-intensity surface fire behavior stabilized quickly because the organic soil could not sustain the fire or burn underground. Seasonality differed between the two experiences, and there was a significant change from February 2023 to March 2023, moving to more adverse meteorological conditions (SMC 2023). However, the range of results could be accommodated according to Canejan's FMA.

Illustrations about technical sheets can be found in the supplementary information (Canejan's FMA example), together with Canejan.

Discussion

The adoption of a transdisciplinary approach in the SFMP of the Aran Valley enabled the integration of criteria and knowledge from local managers, fire experts, and political representatives to co-design an integrated solution. As a result, the strategic framework has been fed by research knowledge and evidence and adopted with



Fig. 7 Capture **a** shows a general perspective of the Canejan managed wildfire on the heel-right flank, visioned from the lookout position. Capture **b** shows the anchoring used to confine the fire to its lower boundary, according to the Canejan's FMA. Capture **c** corresponds to 4 months after the fire (spring). (Credits: GRAF team, CFRS)

political, legal, and executive implications for the entire Aran community. The official approval marks the beginning of a regional paradigm shift in sustainability policies, overcoming conventional fully aggressive suppression strategies and considering solutions more aligned with beneficial ecological fire (Kobziar et al. 2024) and cultural knowledge (Rius et al. 2012; (Fernández-Giménez 2015).

Implementing an EFF policy implies coexisting with fire and co-working with a critical ecological process, building a dynamic balance between suppression (applied when and where negative impacts by wildfires over the common goods prevail) and the use of wildland fire (applied when and where positive impacts over the common goals prevail). It also becomes a cost-effective tool to advance toward fire resilience, offering a space–time window where emergency managers can use a wildfire incident to fulfill land management goals. In the Pyrenees, such a policy involves a meaningful change in the rules for arranging the conventional response to natural disturbances, especially for EMO. The strategy is no longer encapsulated by default within a culture of contingency but accepting fire as an ecological process and managing it proactively to avoid collapse at different levels (Castellnou et al. 2019). This step is critical regarding risk management, especially concerning the low resilience of EMO in the face of simultaneity. With this approach, the current management aligns with the National Association of Forest Service Retirees (WFFP 2022), which states, "Some objectives and situations call for full and aggressive suppression. Others call for minimal on-the-ground actions. Multiple objectives can be achieved on a single fire, meaning that a single fire can be managed as a suppression action but still be shifted into other response tactics on all or part of it to obtain resource benefits or the reverse."

The Canejan case is a fire case in particular example of how multiple concurrent objectives can be achieved on a single fire: tactics to protect the village, fire operations to accompany the fire within the limits of its pre-established FMA, and monitoring the spread of the head of the fire

at its upper limit. Nonetheless, we must remember that fire use is not without risk; in this case, the risk consisted of wildfire spreading beyond the planned area and threatening communities. There are examples of managed wildfires escaping planned boundaries (e.g., the 2009 Mill Flat Fire in Utah, and the 2017 Bonita Fire in New Mexico (Davis 2022). However, the difficulty in identifying and tracking managed wildfires means an exhaustive list of escapes is unavailable. In Europe, the cases are null because of the lack of implementation of this strategy. Although Portugal has some interesting approaches that allow pastural fires to burn (Oliveira et al. 2023), and, more recently, since 2021, the legislation in this country is open to the use of wildfire for managing resources, the explicit practice of wildfire use under technical and strategic decisions is still away from common practices in the continent. Given that aggressive fire suppression may exacerbate fire regimes, along with the impacts of climate change and fuel accumulation (Kreider et al. 2024), adopting a minimum level of uncertainty for wildfire management appears to be a rational approach. Such a shift will require careful risk management, akin to conventional emergency management, and clear communication with the public. However, these challenges should not be seen as barriers; instead, they represent logical and necessary steps toward enhancing a restorative and cost-effective landscape management tool.

It must be pointed out that EFF does not distinguish between different origins of fire. Beyond the type of ignition, managing wildfires differs from implementing prescribed fires in many senses. For example, environmental conditioning and the range of results are more accurate in a prescribed fire environment than during emergency management, mainly because managing an incident involves a different decision-making process than implementing prescribed burns. However, their respective processes have commensurable time–space traceability, implying that both ways of fire use can share metrics and variables and, thus, have common units. This nuance is crucial for fire managers because although their occurrence is radically different, according to the SFMP, both are compatible and can be concurrent when the opportunity arises. We omitted to discuss economic connotations because we do not have data to assess the efficiency of public expenditures. Still, we agree that this topic should be considered soon to assess the economic sustainability of it.

The Canejan fire case integrated pre-established land management goals as tactical objectives and targets into an incident action plan. As far as we know, this case represents the first official register in Europe where managing wildfire strategy has been formally assumed within a fire response environment. Beyond this first experience in Canejan, the achievements obtained in the Aran during the 10-year implementation need more years to be adequately assessed; therefore, the plan's goals require additional time to evaluate its success. However, we can assess the first 2 years of implementation. Beyond that significant step in Canejan, we must also note that the first steps in prescribed fire have also been taken during the first 2 years of the SFMP implementation in different FMAs. In February 2023, the prescribed burn within the FMA of Bossost signifies and accounts for the starting point of the SFMP in terms of prescribed fire. Both events (prescribed fire in Bossost and the managed wildfire in Canejan) are consistent enough to compare the scenario before and after the SFMP approval and determine that the three initially mentioned main objectives related to co-creating this strategic framework are temporally coherent with real experience:

- Firstly, the SFMP existence delivered certainty about the response scenario when unplanned ignitions arose. For certainty about the response scenario, we refer to the state of being confident about how to proceed efficiently and with a high degree of reliability in the face of any unplanned ignition in the Aran, assessing land management goals into emergency decision-making. This has been manifested with minor fire incidents since 2022, the year of approval. Additionally, unplanned ignitions have decreased significantly due to the transversal commitment between the stakeholders involved. This is not a guarantee for the future, but it responds to the level of agreement achieved (unanimously approved by all municipalities of the Aran) for the next 10 years.
- Secondly, the Canejan case noted that for 14 days, the GRAF team was only in place for 4 days as the fire progressed. Once the wildfire was contained within the pre-established FMA, the GRAF team could respond to other concurring incidents in Catalonia (NE Spain), where the CFRS operates. This fact means that the pre-planned strategic scenario delivers the opportunity to allocate the most valued resources according to overall priorities in case of simultaneity.
- Thirdly and finally, both actions achieved surfaces of tenths to hundreds of ha, thus providing an avenue for effective management at the landscape scale.

Conclusions

This study translates recent research efforts on the potential beneficial role of fire in landscape management into a strategic and operational framework providing actions for better adaptation to fire regime shifts in a changing climate. Supported by transdisciplinary

analysis and aimed at action, our case study research focuses on creating a management path toward landscape resilience at social and ecological levels. This fire management implementation path in the Aran can reduce uncertainty in decision-making by proactively anticipating plausible future scenarios. In addition, the subjacent strategic framework dissolves the trap linked to fully aggressive suppression strategies, proposing a transformative role for EMO to overcome the burden of extinction everywhere, every time. Such strategic innovation allows public managers of the Aran to reconcile interests between the stakeholders involved and, simultaneously, to achieve positive feedback with ecosystem services and multifunctional management at the landscape scale.

Addressing fire complexity through transdisciplinary knowledge is a remarkable contribution and recognition of cultural fire's historical role in the Pyrenees' geography. This strategic loop recognizes fire as part of the Pyrenees' cultural heritage and a ubiquitous ecological process linked to their landscape dynamics. Finally, this research attempts to function as a regional catalyst for fire resilience and to inspire other mountain areas to tackle similar challenges in the face of global change trends.

Abbreviations

CFRS	Catalan Fire and Rescue Service
EFF	Environmental flow of fire
EMO	Emergency management organizations
EPP	Emergency preplanning
FMP	Fire management plan
SFMP	Strategic Fire Management Plan of the Aran
FMA	Fire management area
FMU	Fire management unit
GRAF	Wildland fire specialists integrated into the CFRS
IAP	Incident action plan
ICS	Incident command system
Rx	Prescribed fire

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s42408-024-00348-x.

Additional file 1. Supplementary Figure 1. Illustration of common general data linked to FMA technical sheets and, specifically, the FMA of Canejan. It defines the approved pre-planned scenarios, differentiating between prescribed burning, managed wildfire, or both ways in a compatible manner, among other information. Source: SFMP. Supplementary Figure 2. Illustration of axes of confinement linked to the FMA of Canejan. Source: SFMP. Supplementary Figure 3. Illustration of operational recommendations and plot conditionings linked to the FMA of Canejan and, specifically, to the unit known as Hont Hereda. Source: SFMP. Supplementary Figure 4. Illustration of operational recommendations and plot conditionings linked to the FMA of Canejan and, specifically, to the unit known as Era Tueta. Source: SFMP. Supplementary Figure 5. Illustration of EFF prescription explicitly defined at a level unit, in this case, corresponding to the unit Hont Hereda located within the FMA of Canejan. Additionally, it shows the range of acceptable results for this unit, Hont Hereda. Source: SFMP.

Acknowledgements

A team of people with different areas of expertise made this study possible. Firstly, we appreciate the overall support the GRAF Team (wildland fire experts of the CFRS) provided. We especially thank J. Pagès and C. Gallardo for their fieldwork during the initial steps of the project, as well as the generosity and helpful support of J. Pagès in structuring the GIS project. Secondly, we thank the public managers of the Aran for their strategic vision, willingness, open mindset, and local knowledge; M. Bosch and E. Viló from the Aran Fire and Rescue Service; J. Calvo from the Civil Protection of the Aran; G. Arjó and I. Afonso from the Forests and Environment Service of the Aran. All of them contributed to the elaboration of the FMA by transferring their local knowledge, collecting social demands, and providing their expertise and critical geodata to build the GIS project. Finally, we thank the whole community of the Aran for their patience, fire culture, and generosity, clearly demonstrated by the nine existing municipalities of the Aran when unanimously approving the SFMP derived from our applied research work. In addition, MCIN/AEI partly supported the elaboration of this work through the project RESFIRE PID2023-152690OB-C21.

Authors' contributions

JO, MC, and XC provided transdisciplinary knowledge and practical fire expertise. JO outlined the team's organization and wrote the manuscript, and AD, MR, and LB helped structure the manuscript and provided extensive editorial input and citations during manuscript preparation and revision. JO proposed, and all authors developed, the idea of the environmental flow of fire (EFF) as a standard for fire management, linking preexisting ideas, such as emergency preplanning or managing wildfire, creatively. MC, XC, and JO proposed the creative niche emergency management organizations can play to better adapt to fire regime shifts forced by global change and the idea of resilience for emergency management organizations, which materialized through the capability of triage to deal with simultaneity, focusing on making the SFMP credible at a tactical implementation level. JO and XC are the leading developers of the FMA, and they also built the GIS project. AD, LB, MR, and JO (this one as part of his PhD) provided scientific skills regarding fire regime characterization and landscape modeling assessment. Finally, all authors read and approved the final manuscript.

Funding

No additional or outside funding was used to develop this manuscript.

Data availability

The datasets used and analyzed during the current study are available from the corresponding author upon reasonable request. Material relative to the Medfire Aran Model is available at https://github.com/AndreaDuane/MedfireAran.

Declarations

Ethics approval and consent to participate.

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare that they have no competing interests.

Author details

[1]University Research Institute for Sustainability Science and Technology, Universitat Politècnica de Catalunya – BarcelonaTech, Jordi Girona 31, Barcelona 08034, Spain. [2]Catalan Fire and Rescue Service (CFRS), GRAF Team. Avinguda de Serra Galliners, Universitat Autònoma de Barcelona), Cerdanyola del Vallès 08193, Spain. [3]Department of Agricultural and Forest Engineering, Universitat de Lleida, Alcalde Rovira Roure 191, Lleida 25198, Spain. [4]Forest Science Centre of Catalonia (CTFC), Carretera Vella de Sant Llorenç de Morunys Km 2, Solsona, Lleida 25280, Spain. [5]Ecological and Forestry Applications Research Center (CREAF), Campus de Bellaterra (UAB) Edifici C, Cerdanyola del Vallès 08193, Spain. [6]The Spanish National Research Council (CSIC), Cerdanyola del Vallès 08193, Spain. [7]Department of Environmental Science and Policy, University of California, Davis, Davis, 95616, USA.

Received: 26 June 2024 Accepted: 27 December 2024
Published online: 14 February 2025

References

Agee, James K. 1998. The Landscape Ecology of Western Forest Fire Regimes. *Northwest Science* 72 (Special Issue): 24–34.

Alvarez, A., J. Lecina-Diaz, E. Batllori, A. Duane, L. Brotons, and J. Retana. 2024. Spatiotemporal patterns and drivers of extreme fire severity in Spain for the period 1985–2018. *Agricultural and Forest Meteorology* 358 (July): 110185. https://doi.org/10.1016/j.agrformet.2024.110185.

Amblar-Francés, María P., Petra Ramos-Calzado, Jorge Sanchis-Lladó, Alfonso Hernanz-Lázaro, María C. Peral-García, Beatriz Navascués, Marta Dominguez-Alonso, María A. Pastor-Saavedra, and Ernesto Rodríguez-Camino. 2020. High Resolution Climate Change Projections for the Pyrenees Region. *Advances in Science and Research* 17: 191–208. https://doi.org/10.5194/asr-17-191-2020.

Arilla, Etel, Xavier Castellarnau, Marc Castellnou, Jordi Castellví, Jose Cespedes, Laia Estivill, Aniol Ferragut, Asier Larrañaga, Marta Miralles, Edgar Nebot, Jordi Pagès, Pepe Pallàs, Martí Rosell, and Borja Ruiz. 2023. Piloting the adaptation of the methodology of Forest Fire Potential Polygons to improve decision making on EWE events. Zenodo. .

Arno SF, 1994. "Fire Ecology of Pacific Northwest Forests, by J.K. Agee." International Journal of Wildland Fire. https://doi.org/10.1071/wf9940195.

Bal, M.C., A. Pelachs, R. Perez-Obiol, et al. 2011. Fire history and human activities during the last 3300cal yr BP in Spain's Central Pyrenees: The case of the Estany de Burg. *Palaeogeography, Palaeoclimatology, Palaeoecology* 300: 179. https://doi.org/10.1016/j.palaeo.2010.12.023.

Bean, Rachel, and Alexander Evans. 2023. Managed Wildfire: A Research Synthesis & Overview, no. April: 15. https://www.swfireconsortium.org/2023/04/18/managed-wildfire/.

Brotons L, Aquilué N, de Cáceres M, et al. (2013). How Fire History, Fire Suppression Practices and Climate Change Affect Wildfire Regimes in Mediterranean Landscapes. PLoS One 8:. https://doi.org/10.1371/journal.pone.0062392

Buisson, E., S. Archibald, and A. Fidelis. 1979. Suding KN (2022) Ancient grasslands guide ambitious goals in grassland restoration. *Science* 377: 594–598. https://doi.org/10.1126/SCIENCE.ABO4605.

Castellnou, M., N. Prat-Guitart, E. Arilla, et al. 2019. Empowering strategic decision-making for wildfire management: avoiding the fear trap and creating a resilient landscape. *Fire Ecol* 15: 1.

Servei Meteorologic de Catalunya, 2023. Butlletí Anual d'Indicadors Climàtics. Generalitat de Catalunya. Link: https://www.meteo.cat/wpweb/climatologia/butlletins-i-episodis-meteorologics/butlleti-anual-dindicadors-climatics/

Copes-Gerbitz, K., S.M. Hagerman, and L.D. Daniels. 2022. Transforming fire governance in British Columbia, Canada: An emerging vision for coexisting with fire. *Regional Environmental Change* 22: 1–15. https://doi.org/10.1007/S10113-022-01895-2/FIGURES/3.

Costa, Pau, Marc Castellnou, Asier Larrañaga, Marta Miralles, Daniel Kraus, Pau Costa Alcubierre, Marc Castellnou, et al. 2011. Prevention of Large Wildfires Using the Fire Types Concept. https://www.researchgate.net/publication/263923019_Prevention_of_Large_Wildfires_using_the_Fire_Types_Concept.

Coughlan, Michael R. 2014. Farmers, flames, and forests: historical ecology of pastoral fire use and landscape change in the French Western Pyrenees, 1830–2011. *Forest Ecology and Management* 312: 55–66. https://doi.org/10.1016/j.foreco.2013.10.021.

Cramer, Wolfgang, Joël Guiot, Katarzyna Marini, Brian Azzopardi, Mario V Balzan, Semia Cherif, Enrique Doblas-Miranda, et al. 2020. MedECC 2020 Summary for Policymakers. Climate and Environmental Change in the Mediterranean Basin – Current Situation and Risks for the Future. First Mediterranean Assessment Report, November. https://doi.org/10.5281/ZENODO.5513887.

Davim, D.A., C.G. Rossa, J.M.C. Pereira, and P.M. Fernandes. 2022. Evaluating the effect of prescribed burning on the reduction of wildfire extent in Portugal. *SSRN Electronic Journal* 519: 120302. https://doi.org/10.2139/ssrn.4046582.

Davis EJ, Huber-Stearns H, Caggiano M, McAvoy D, Cheng AS, Deak A, Evans A, 2022. Managed wildfire: a strategy facilitated by civil society partnerships and interagency cooperation. Society and Natural Resources. 1–19. https://doi.org/10.1080/08941920.2022.2092803

de V, Resco Dios, J. Hedo, A. Cunill Camprubí, P. Thapa, E. Del Martínez Castillo, J. de Martínez Aragón, J.A. Bonet, et al. 2021. Climate change induced declines in fuel moisture may turn currently fire-free pyrenean mountain forests into fire-prone ecosystems. *Science of the Total Environment* 797: 149104. https://doi.org/10.1016/j.scitotenv.2021.149104.

Rigo, Daniele de, Giorgio Libertà, Tracy Houston Durrant, Tomàs Artés Vivancos, and Jesús San-Miguel-Ayanz. 2017. Forest Fire Danger Extremes in Europe under Climate Change: Variability and Uncertainty. Publication Office of the European Union. https://doi.org/10.2760/13180.

Duane, A., N. Aquilué, A. Gil-Tena, and L. Brotons. 2016. Integrating fire spread patterns in fire modeling at the landscape scale. *Environmental Modelling & Software* 86: 219–231. https://doi.org/10.1016/j.envsoft.2016.10.001.

Duane, A., N. Aquilué, Q. Canelles, et al. 2019. Adapting prescribed burns to future climate change in Mediterranean landscapes. *Science of the Total Environment*. https://doi.org/10.1016/j.scitotenv.2019.04.348.

Duane, A., M. Castellnou, and L. Brotons. 2021. Towards a comprehensive look at global drivers of novel extreme wildfire events. *Climate Change.* 165: 43. https://doi.org/10.1007/s10584-021-03066-4.

Duane, Andrea, Lluís Brotons, Jordi Oliveres, i Núria Aquilué. «El Model MED-FIRE Val D'aran. Disseny, Inicialització, Simulació i Resultats.». Zenodo, 1 juliol 2020. https://doi.org/10.5281/zenodo.11238569.

Fernandes, P.M. 2013. Fire-Smart management of forest landscapes in the Mediterranean basin under global change. *Landscape and Urban Planning.* 110: 175. https://doi.org/10.1016/j.landurbplan.2012.10.014.

Fernandes, P.M., G.M. Davies, D. Ascoli, et al. 2013. Prescribed burning in southern Europe: Developing fire management in a dynamic landscape. *Frontiers in Ecology Environment* 11: e4.

Fernandes, Paulo M., C.G. Rossa, J. Madrigal, E. Rigolot, D. Ascoli, C. Hernando, N.G. Guiomar, and M. Guijarro. 2022. "Prescribed Burning in the European Mediterranean Basin." Global Applications of Prescribed Fire., 230–48. https://www.publish.csiro.au/book/7909/.

Fernández-Giménez ME, 2015. "A Shepherd Has to Invent: Poetic Analysis of Social-Ecological Change in the Cultural Landscape of the Central Spanish Pyrenees." Ecology and Society 20 (4). https://doi.org/10.5751/ES-08054-200429.

Fire Management Board. 2019. Federal Wildland Fire Policy Terms and Definitions, no. October: 1. https://fs-prod-nwcg.s3.us-gov-west-1.amazonaws.com/s3fs-public/2023-06/eb-fmb-m-19-004a.pdf?VersionId=NKG9pNAhwyTbrRjL43AX2K0xyMpnWwjz.

Fry, D.L., S.L. Stephens, B.M. Collins, et al. 2014. Contrasting spatial patterns in active-fire and fire-suppressed Mediterranean climate old-growth mixed conifer forests. *PLoS ONE*. https://doi.org/10.1371/journal.pone.0088985.

González-Guerrero, O., and X. Pons. 2020. The 2017 Land Use/Land cover map of Catalonia based on sentinel-2 images and auxiliary data. *Revista De Teledetección* 2020 (55): 81–92. https://doi.org/10.4995/raet.2020.13112.

González-Hidalgo, M., I. Otero, and G. Kallis. 2014. Seeing beyond the smoke: The political ecology of fire in Horta de Sant Joan (Catalonia). *Environment and Planning A.* 46: 1014. https://doi.org/10.1068/a45600.

Kobziar, L.N., J.K. Hiers, C.M. Belcher, W.J. Bond, C.A. Enquist, E.L. Loudermilk, J.R. Miesel, et al. 2024. Principles of Fire Ecology. *Fire Ecology* 20 (1): 39. https://doi.org/10.1186/s42408-024-00272-0.

Kreider, R., P.E. Higuera, S.A. Parks, N. White, A.J. Larson, and W.L. Rice. 2024. *Fire suppression makes wildfires more severe and accentuates impacts of climate change and fuel accumulation.* 15: 2412. https://doi.org/10.1038/s41467-024-46702-0.

Loepfe, L., J. Martínez-Vilalta, J. Oliveres, et al. 2010. Feedback between fuel reduction and landscape homogenization determines fire regimes in three Mediterranean areas. *Forest Ecology and Management* 259: 2366. https://doi.org/10.1016/j.foreco.2010.03.009.

Messana, C., C. Tornero, R. Madgwick, A.L. Lamb, J. Evans, and L. Colominas. 2023. Between Valleys, Plateaus, and Mountains: Unveiling livestock altitudinal mobility in the iron Age Iberian Peninsula (3rd c. BC) through a multi-isotope approach. *Frontiers in Environmental Archaeology* 2: 1245725. https://doi.org/10.3389/fearc.2023.1245725.

Minnich RA (1983) Fire mosaics in southern California and northern Baja California. Science (1979). https://doi.org/10.1126/science.219.4590.1287

Montiel, C., and D. T. Kraus. 2010. Best practices of fire use: prescribed burning and suppression: fire programmes in selected case-study regions in Europe. Porvoo: European Forest Institute.

Moreira, F., D. Ascoli, H. Safford, et al. 2020. Wildfire management in Mediterranean-type regions: Paradigm change needed. *Environmental Research Letters* 15: 011001.
Newman Thacker, F.E., M.C. Ribau, H. Bartholomeus, and C.R. Stoof. 2023. What is a fire resilient landscape? Towards an integrated definition. *Ambio* 52: 1592–1602. https://doi.org/10.1007/s13280-023-01891-8.
Oliveira, E., and P.M. Fernandes. 2023. Pastoral burning and its contribution to the fire regime of Alto Minho Portugal. *Fire* 6 (5): 210. https://doi.org/10.3390/fire6050210.
Oliveres, J., X. Castellarnau, G. Arjó, and E. Viló. 2023. Canejan Managed Fire, 03/16/2023. https://interior.gencat.cat/web/.content/home/030_arees_dactuacio/bombers/foc_forestal/consulta_incendis_forestals/informes_incendis_forestals/2020-2029/2023/20230316_VA_Canejan-Managed-Fire-report.pdf.
Otero, I., and J. Nielsen. 2017. Coexisting with wildfire? Achievements and challenges for a radical social-ecological transformation in Catalonia (Spain). *Geoforum* 85: 234. https://doi.org/10.1016/j.geoforum.2017.07.020.
Otero, I., M. Castellnou, I. González, E. Arilla, L. Castell, J. Castellví, F. Sánchez, and J. O. Nielsen. 2018. Democratizing wildfire strategies. Do you realize what it means? Insights from a participatory process in the Montseny Region (Catalonia, Spain). *PLoS ONE* 13:e0204806.
Pastor, E., Y. Pétrez, M. Miralles, and E. Planas. 2009. Prescribed Burning in Catalonia: Fire Management and Research. *Proceedings of the Royal Society of Queensland* 115 (April): 23–27. https://doi.org/10.5962/p.357722.
Pausas, J.G. 2019. Generalized fire response strategies in plants and animals. *Oikos*. https://doi.org/10.1111/oik.05907.
Pausas, J.G., and J.E. Keeley. 2019. Wildfires as an ecosystem service. *Frontiers in Ecology and the Environment*. https://doi.org/10.1002/fee.2044.
Rego, F., E. Rigolot, P. Fernandes, et al. 2010. Towards integrated fire management. *EFI Policy Brief* 4: 16.
Rius, D., B. Vannière, and D. Galop. 2012. Holocene History of Fire, Vegetation and Land Use from the Central Pyrenees (France). *Quaternary Research* 77 (1): 54–64. https://doi.org/10.1016/j.yqres.2011.09.009.
Silva, Joaquim, Rego, Francisco, Fernandes, Paulo, and Rigolot, Eric. 2010. Towards Integrated Fire Management—Outcomes of the European Project Fire Paradox.
Scanlon J, 2003. Flow: The essentials of environmental flows. flow: the essentials of environmental flows. https://doi.org/10.2305/iucn.ch.2003.wani.2.en.
Schoennagel, T., J.K. Balch, H. Brenkert-Smith, et al. 2017. Adapt to more wildfire in western North American forests as climate changes. *Proc Natl Acad Sci U S A* 114: 4582–4590. https://doi.org/10.1073/pnas.1617464114.
Maria Vergés Pérez. 2022. Strategic Fire Management Plan of the Aran. https://tauler.seu.cat/pagDetall.do?idEdicte=354679&idens=8103980001
Tedim, F.V., L.M. Amraoui, et al. 2018. Defining extreme wildfire events: difficulties, challenges, and impacts. *Fire* 1: 9. https://doi.org/10.3390/fire1010009.
Tickner, D., H. Parker, C.R. Moncrieff, N.E.M. Oates, E. Ludi, and M. Acreman. 2017. Managing rivers for multiple benefits-a coherent approach to research, policy and planning. *Frontiers in Environmental Science*. 5: 4. https://doi.org/10.3389/fenvs.2017.00004.
Tölgyesi, C., E. Buisson, A. Helm, et al. 2022. Urgent need for updating the slogan of global climate actions from "tree planting" to "restore native vegetation." *Restoration Ecology* 30: e13594. https://doi.org/10.1111/rec.13594.
Wagtendonk,. 2007. The History and Evolution of Wildland Fire Use. *Fire Ecology*. https://doi.org/10.4996/firecology.0302003.
Wildland, Federal, and Fire Policy. 2022. Managing Wildland Fire A Position of the National Association of Forest Service Retirees Position: Issue: no. February: 1–6. https://nafsr.org/committies/fire/022422%20Managing%20Wildland%20Fire%20Position%20Paper.pdf.
Xanthopoulos, Gavriil, Vittorio Leone, and Giuseppe Mariano Delogu. 2020. The Suppression Model Fragilities: The 'Firefighting Trap. In Extreme Wildfire Events and Disasters: Root Causes and New Management Strategies. Elsevier. Pages 135-153, ISBN 9780128157213. https://doi.org/10.1016/B978-0-12-815721-3.00007-2.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.